

2.) přesečí 3 čar

a) p: $\vec{r} = (1, 2, 0) + t(1, 1, 1)$

q: $\frac{x-2}{2} = \frac{y-3}{3} = 1-z$

$\vec{r}_0 = (1, 2, 0)$

$\vec{r}'_0 = (2, 3, -1)$

$\vec{r} = (1, 2, 0) + t(1, 1, 1) = (2, 3, 1) + t'(2, 3, -1)$

$1+t = 2+2t' \quad 2+t = 3+3t' \quad t = 1-t'$

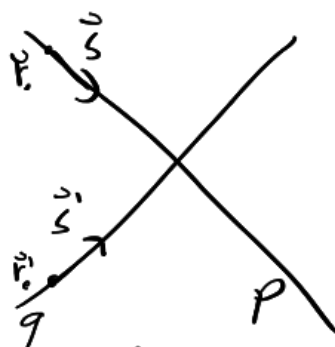
$2-t' = 2+2t'$

$3t' = 0$

$t' = 0$

$t = 1$

$A = (2, 3, 1) + 0 \cdot (2, 3, -1) = \underline{\underline{(2, 3, 1)}}$



b)

$\pi: 2x + 3y - z + 1 = 0$

$\vec{n}_\pi \times \vec{n}_\Sigma = \begin{vmatrix} i & j & k \\ 2 & 3 & -1 \\ 1 & -1 & 1 \end{vmatrix} \begin{vmatrix} i & j \\ 2 & 3 \\ 1 & -1 \end{vmatrix}$

$\Sigma: x - y + z = -8$

$= 3i - j - 2k - 3k - i - 2j =$

$= (2, -3, -5) = \vec{s}$

$x = 0:$

$3y - z = -1$

$-y + z = 8$

$2y = 7 \quad z = 8 + y = \frac{23}{2}$

$y = \frac{7}{2}$

$\vec{r} = (0, \frac{7}{2}, \frac{23}{2}) + t(2, -3, -5)$

4.

$A = (3, 4, 1)$

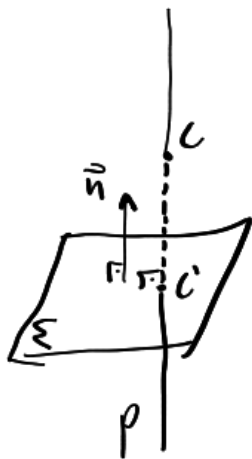
$B = (-1, 0, 5)$

$\vec{r}_0 = (1, 2, 3)$

$\vec{s} = (1, 1, 1)$

$$C = (6, 5, -4)$$

enako oddaljene od A in B,
najbližje C



$$AB = (-4, -1, 1)$$

$$\vec{n} = (-1, -1, 1)$$

$$\sum: ((1, 2, 3) - (x, y, z)) \cdot (1, 1, -1) = 0$$

$$(1-x, 2-y, 3-z) \cdot (1, 1, -1) = 0$$

$$1-x+2-y-3+z=0$$

$$-x-y+z=0$$

$$x+y-z=0$$

$$p: \vec{r} = (6, 5, -4) + t(-1, -1, 1) = (x, y, z)$$

$$x = 6-t$$

$$y = 5-t$$

$$z = -4+t$$

$$6-t+5-t+4-t=0$$

$$3t = 15$$

$$t = 5$$

$$C' = (6, 5, -4) + 5(-1, -1, 1)$$

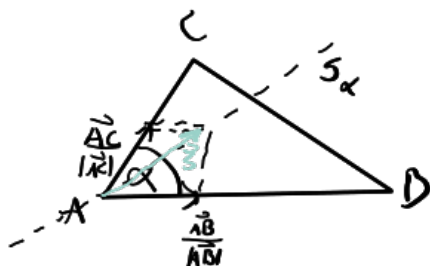
$$\underline{\underline{C' = (1, 0, 1)}}$$

6. $A = (1, -1, -1)$

$$B = (3, 1, 0)$$

$$C = (-3, 3, -3)$$

$$s_x = ?$$



$$s_x: \vec{r} = \vec{r}_0 + t\vec{s}$$

$$\vec{r}_0 = (1, -1, 1)$$

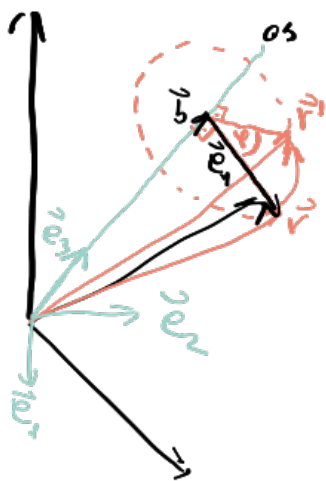
$$\vec{s} = \frac{\vec{AB}}{|\vec{AB}|} + \frac{\vec{AC}}{|\vec{AC}|} = \frac{(2, 2, 1)}{\sqrt{4+4+1}} + \frac{(-4, 4, -2)}{\sqrt{36}}$$

$$= \frac{(2, 2, 1)}{3} + \frac{(-4, 4, -2)}{6} = \left(\frac{2}{3} - \frac{2}{3}, \frac{2}{3} + \frac{2}{3}, \frac{1}{3} - \frac{1}{3}\right) =$$

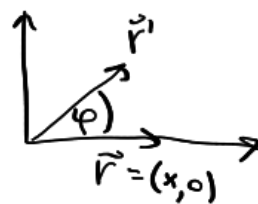
$$= \left(0, \frac{4}{3}, 0\right)$$

$$\underline{\underline{\vec{r} = (1, -1, 1) + t(0, \frac{4}{3}, 0)}}$$

7. Formula za rotacijo za kot φ okoli osi v prostoru:



v ravnini:



$$\vec{r}' = x \cdot (\cos \varphi, \sin \varphi)$$

Podane: $\vec{r}, \vec{e}, \varphi$
 $|\vec{e}| = 1$

$$\vec{e}_3 = \vec{e}$$

$$\vec{b} = \left(\vec{r} \cdot \frac{\vec{e}}{|\vec{e}|} \right) \cdot \frac{\vec{e}}{|\vec{e}|} = (\vec{r} \cdot \vec{e}) \cdot \vec{e}$$

$$\vec{e}_1 = \frac{-\vec{b} + \vec{r}}{|\vec{r} - (\vec{r} \cdot \vec{e}) \vec{e}|} = \frac{\vec{r} - (\vec{r} \cdot \vec{e}) \vec{e}}{\sqrt{|\vec{r}|^2 - (\vec{r} \cdot \vec{e})^2}}$$

$$|\vec{r} - (\vec{r} \cdot \vec{e}) \vec{e}| = \sqrt{|\vec{r}|^2 - 2\vec{r} \cdot (\vec{r} \cdot \vec{e}) \vec{e} + ((\vec{r} \cdot \vec{e}) \vec{e}) \cdot ((\vec{r} \cdot \vec{e}) \vec{e})} =$$

$$= \sqrt{|\vec{r}|^2 - 2(\vec{r} \cdot \vec{e})^2 + (\vec{r} \cdot \vec{e})^2} = \sqrt{|\vec{r}|^2 - (\vec{r} \cdot \vec{e})^2}$$

$$\vec{e}_2 = \frac{\vec{e} \times \vec{r} - \vec{e} \times ((\vec{r} \cdot \vec{e}) \vec{e})}{\sqrt{|\vec{r}|^2 - (\vec{r} \cdot \vec{e})^2}} = \frac{\vec{e} \times \vec{r}}{\sqrt{|\vec{r}|^2 - (\vec{r} \cdot \vec{e})^2}}$$

$$\vec{r} = A \vec{e}_1 + B \vec{e}_2 + C \vec{e}_3 = (\vec{r} \cdot \vec{e}_1) \vec{e}_1 + (\vec{r} \cdot \vec{e}_2) \vec{e}_2 + (\vec{r} \cdot \vec{e}) \vec{e} =$$

$$= \frac{|\vec{r}|^2 - (\vec{r} \cdot \vec{e})^2}{\sqrt{|\vec{r}|^2 - (\vec{r} \cdot \vec{e})^2}} \cdot \vec{e}_1 + \frac{(\vec{e} \times \vec{r}) \cdot \vec{r}}{\sqrt{|\vec{r}|^2 - (\vec{r} \cdot \vec{e})^2}} \cdot \vec{e}_2 + (\vec{r} \cdot \vec{e}) \vec{e} =$$

$$= \sqrt{|\vec{r}|^2 - (\vec{r} \cdot \vec{e})^2} \cdot \vec{e}_1 + (\vec{r} \cdot \vec{e}) \vec{e}$$

$$\vec{r}' = A \cos \varphi \vec{e}_1 + A \sin \varphi \vec{e}_2 + (\vec{r} \cdot \vec{e}) \vec{e} =$$

$$= \sqrt{|\vec{r}|^2 - (\vec{r} \cdot \vec{e})^2} \cos \varphi \vec{e}_1 + \sqrt{|\vec{r}|^2 - (\vec{r} \cdot \vec{e})^2} \sin \varphi \vec{e}_2 + (\vec{r} \cdot \vec{e}) \vec{e} =$$

$$= (\vec{r} - (\vec{r} \cdot \vec{e}) \vec{e}) \cos \varphi + (\vec{e} \times \vec{r}) \sin \varphi + (\vec{r} \cdot \vec{e}) \vec{e} =$$

$$\vec{r}' = \cos \varphi \vec{r} + (\vec{r} \cdot \vec{e}) \vec{e} (1 - \cos \varphi) + (\vec{e} \times \vec{r}) \sin \varphi \quad \text{Rodriguezova formula}$$

8. $p: \vec{r}(t) = \overbrace{(2, 3, 1)}^A + t(1, 1, 1)$
 $T = (5, 5, 3)$

a) π dolozena z p, T ?

$$(\vec{r} - \vec{r}_0) \cdot \vec{n} = 0$$

$$\vec{n} = \vec{A} \times \vec{s} = \begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 3 & 2 & 2 \end{vmatrix} = 2i + 2j + 3k - 2i - 3j - 2k = -j + k = (0, -1, 1)$$

$$(\vec{r} - (5, 5, 3)) \cdot (0, -1, 1) = 0$$

$$(x-5, y-5, z-3) \cdot (0, -1, 1) = 0$$

$$\boxed{\pi: z - y = -2}$$

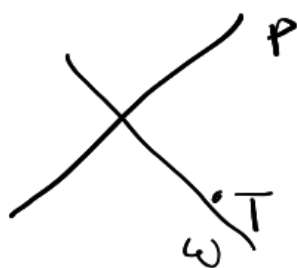
$$d(\pi, (0, 0, 0)) = ?$$

$$q: \vec{r} = t'(0, -1, 1)$$

$$\begin{array}{ll} \swarrow & \\ x=0, & z-y=-2 \\ y=-t' & t'+t'=-2 \\ z=t' & t'=-1 \end{array}$$

$$d((0, 1, -1), (0, 0, 0)) = \sqrt{1+1} = \underline{\underline{\sqrt{2}}}$$

b) $T \in \omega \in \pi; \omega \perp p$
 $\omega = ?$



$$\omega: \vec{r} = (5, 5, 3) + t \cdot \vec{s}$$

$$\vec{s} = (0, -1, 1) \times (1, 1, 1) = \begin{vmatrix} i & j & k \\ 0 & -1 & 1 \\ 1 & 1 & 1 \end{vmatrix} =$$

$$= -i + j + 0k - i + k = (-2, 1, 1)$$

$$\boxed{\omega: \vec{r} = (5, 5, 3) + t(-2, 1, 1)}$$

$$I(x_0, y_0, z_0) \quad \Pi: ax+by+cz=d$$

$$d(T, \Pi) = \frac{|ax_0+by_0+cz_0-d|}{\sqrt{a^2+b^2+c^2}}$$

9.

$$p: \vec{r} = (-1, 2, 1) + t(-2, 2, -1)$$

$$q: \frac{x}{2} = y-1 = \frac{z-6}{-2} \rightarrow \vec{r} = (0, 1, 6) + t(2, 1, -2)$$

Π enako oddaljena od p, q ju ne sekata

$$\vec{s}_p \times \vec{s}_q = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -2 & 2 & -1 \\ 2 & 1 & -2 \end{vmatrix} = -4\vec{i} - 2\vec{j} - 2\vec{k} + \vec{i} - 4\vec{j} - 4\vec{k} = (-3, -6, -6)$$

$$\vec{r}_0 = \frac{(-1, 2, 1) + (0, 1, 6)}{2} = \left(-\frac{1}{2}, \frac{3}{2}, \frac{7}{2}\right) \quad \vec{n} = (1, 2, 2)$$

$$\Pi: (\vec{r} - \vec{r}_0) \cdot \vec{n} = 0$$

$$\left(x + \frac{1}{2}, y - \frac{3}{2}, z - \frac{7}{2}\right) \cdot (1, 2, 2) = 0$$

$$x + \frac{1}{2} + 2y - 3 + 2z - 7 = 0$$

$$\boxed{x + 2y + 2z = \frac{19}{2}}$$

